



Alberta Public Health Disease Management Guidelines

California Serogroup Infection

Includes Jamestown Canyon Virus and Snowshoe Hare Virus



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Alberta Health

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Contents

Case Definition	4
Confirmed Case	4
Probable Case	4
Reporting Requirements	5
Physicians, Health Practitioners and Others	5
Laboratories	5
Alberta Health Services and First Nations and Inuit Health Branch	5
Epidemiology.....	6
Etiology	6
Clinical Presentation.....	6
Diagnosis	6
Treatment.....	6
Reservoir.....	6
Transmission.....	6
Incubation Period	7
Period of Communicability.....	7
Host Susceptibility	7
Incidence.....	7
Public Health Management.....	8
Key Investigation.....	8
Management of a Case.....	8
Management of Contacts	8
Preventive Measures.....	8
Appendix 1: Revision History	9
References.....	10

Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Detection of Jamestown Canyon (JC) or Snowshoe Hare (SSH) viral nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., blood, cerebrospinal fluid [CSF])

OR

- Positive serum IgM antibodies for JC/SSH virus and a seroconversion demonstrated by fourfold or greater rise in JC/SSH virus-specific PRNT antibody titres detected in acute and convalescent-phase sera ideally collected at least two weeks apart^(B) and occurring during a period when and where JC/SSH transmission is likely^(C)

OR

- Presence of JC/SSH IgM in a CSF sample and a PRNT titre of $\geq 1:20$ in acute sera and occurring during a period when and where JC/SSH transmission is likely^(C)

Probable Case

Clinical illness^(A) with the following:

- Positive/equivocal serum IgM and a PRNT titre of $\geq 1:20$ ^(D) for antibodies specific for JC/SSH virus on a single acute phase serum specimen taken during a period when and where JC/SSH transmission is likely^(C)

^(A) Clinical illness is characterized by a febrile illness of variable severity associated with neurological symptoms ranging from headache to aseptic meningitis or encephalitis. Arboviral encephalitis cannot be distinguished clinically from other central nervous system (CNS) infections without laboratory testing. Symptoms can include headache, confusion or other alteration in sensorium, nausea and vomiting. Signs may include fever, meningismus, cranial nerve palsies, paresis or paralysis, sensory deficits, altered reflexes, convulsions, abnormal movements and coma of varying degree.

^(B) Seroconversion indicates a recent infection with an infectious agent such as an arbovirus. For detecting JC and SSH virus, antibody IgM and IgG ELISAs may cross react to some extent. PRNT confirmatory testing documents a specific exposure to these viruses and may be requested at the National Microbiology Laboratory through the Public Health Laboratory (ProvLab). However, due to the possibility of persistent IgM in serum samples, the demonstration of a seroconversion (or detection of virus) is necessary to associate positive serology with current illness.

^(C) The highest risk of JC/SSH exposure occurs mid-July to early fall in Canada and the Midwest United States (US) and year round in southeast parts of the US.⁽¹⁾ Cases of febrile and meningitis/encephalitis have been documented in late spring and early summer.

^(D) IgM in an acute serum sample and a single antibody titre of $\geq 1:20$ by PRNT suggests recent infection. It is recommended that a second serum sample be collected (see above). A second laboratory result with a stable (unchanged/static or $\leq$ twofold rise) antibody titre is still suggestive of recent infection but may depend on when the specimen was taken in relation to onset of symptoms (e.g., the rise in titre may be missed due to timing of sample collection). Note – The presence of JC/SSH specific IgM in an acute CSF specimen is confirmatory.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases of JC/SSH encephalitis in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases of JC/SSH to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two days) including:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

California serogroup (CSG) viruses are members of the *Bunyaviridae* family and include Jamestown Canyon (JC) virus and Snowshoe hare (SSH) virus.⁽²⁾ JC virus was first isolated from a mosquito collected in Jamestown Canyon, Colorado, in 1961.⁽³⁾ SSH was first identified in a snowshoe hare in Montana in 1958.⁽⁴⁾

Clinical Presentation

The majority of infections are likely asymptomatic, similar to West Nile virus (WNV). Symptoms can range from mild with fever, headache and vomiting to severe with high fever, meningeal symptoms, tremors, and occasionally seizures (especially in children); rarely acute flaccid paralysis can occur. Severe infections result in a variety of sequelae such as behaviour changes, learning disabilities, and cognitive deficits.⁽²⁾

Diagnosis

Recent advances in diagnostic testing using both serological and molecular assays have made it possible to more accurately identify acute cases. To diagnose CSG infections, testing of acute and convalescent sera from suspect cases are recommended for determining the diagnostic rise or decrease in CSG specific antibody titres. As well, samples of CSF should be included for detection of acute IgM antibody or viral genomic sequences (PCR), which would constitute confirmatory laboratory evidence of an infection associated with accompanying clinical characteristics.⁽⁵⁾

The dynamics of antibody rise during a CSG infection is similar to WNV. Approximately one week after the onset of symptoms, a full IgM response should be detectable. IgG will begin to be present at about 10 days after onset of symptoms, and variations to this will depend upon the patient. IgG should last several years after exposure. In certain cases there is evidence of persistent IgM (as observed with WNV), which needs to be taken into consideration for diagnostic purposes (Personal communication, Drebot M, National Microbiology Laboratory, 2010 Nov 23).

Treatment

There is no known antiviral therapy available.

Reservoir

The amplifying hosts/reservoirs for CSG viruses can be either small mammals such as squirrels, chipmunks, and hares for SSH viruses, or larger animals such as deer and elk for JC viruses. *Aedes*, *Culex* and *Anopheles* mosquitoes,⁽²⁾ which are present throughout Canada from May to October, allow CSG to overwinter.⁽⁵⁾ These mosquito species feed mainly from dusk until dawn, although some *Aedes* species also feed during the day (Personal communication, McIntosh J, Alberta Environment, 2010 Nov 17).

Transmission

CSG viruses are transmitted to humans through the bite of an infected mosquito. The mosquito becomes infected after feeding on a source infected with the virus and then spreading the infection in subsequent blood meals.⁽²⁾

Although rare, CSG viruses may be transmitted from person to person through blood transfusion, organ transplantation, intrauterine transmission and possibly milk.^(1,6)

Incubation Period

The incubation period of CSG viruses is between five and 15 days.⁽⁷⁾

Period of Communicability

CSG viruses are not directly transmitted from person to person. Mosquitoes are infective for life.⁽⁷⁾

Host Susceptibility

Identified cases of SSH virus have been predominately associated with encephalitis in children.⁽⁵⁾ Although JC virus appears to be primarily associated with illness in adults, cases involving individuals younger than 18 years of age have also been documented.

Incidence

General

CSG viruses are found mainly in Canada and the US, and both JC and SSH have a wide geographic distribution. In the US, JC is found throughout the West and Midwest, although most human cases have been found in New York, New England and the upper Midwest states. JC seroconversion has more recently been documented in local populations in Alaska. SSH is found internationally in Russia, China and Alaska.⁽⁸⁾

Based on serosurveys in Canada and the US, it is estimated that approximately 25% of the population may have antibodies to these viruses. The actual seroprevalence rates may range from 4–50%, depending upon the region.^(5,9,10)

Canada

CSG is not nationally notifiable, so it is difficult to estimate the burden of this disease in Canada. JC has been found in Newfoundland, Quebec, Ontario, Manitoba, Saskatchewan and the Northwest Territories;⁽¹¹⁾ SSH has been documented in all provinces and territories.⁽⁸⁾

In 2009, eight acute cases of exposure were confirmed in Canada, with approximately 100 probable cases based on the presence of virus specific IgM and significant PRNT titres. In the serums of patients evaluated for suspect WNV infections (febrile and neurological illness) in Manitoba between 2008 and 2009, 2% of the patients were identified as having evidence of SSH virus exposures and 9% were identified as having evidence of JC virus infection. Similar detection rates of CSG virus antibody were identified among suspect WNV cases (later determined to be WNV negative) in Ontario and Saskatchewan. Although patients may have SSH or JC specific IgM antibody, it is not known if these results represent current or previous infections, due to the possible persistence of IgM in certain cases.⁽⁵⁾

Alberta

CSG became notifiable in Alberta in 2011. Between 2011 and 2020, 22 cases of both JC (17) and SSH (5) have been reported in Alberta.

Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

Assess potential risk factors and likely mode of transmission for the acquisition of CSG within three weeks prior to onset of symptoms, as well as other considerations:

- living in or travel to an area when and where JC/SSH transmission is present (see Incidence section), and
- being bitten by mosquitoes.

Management of a Case

- Supportive care including IV fluids, ventilation and the prevention of secondary infections.
- Symptomatic treatment.

Management of Contacts

- There is no evidence to suggest that CSG virus can be transmitted to household contacts of infected persons.

Preventive Measures

- Preventing mosquito bites is considered the best measure to avoid the low risk of contracting JC and SSH infection.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links
	Case Definition	• Suspect Case definition removed as it is not notifiable to Alberta Health.
	Incidence	• Updated Alberta incidence and added link to IHDA.

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